Enquiry for Science Major/Minor/Programme Requirements

<table>
<tr><td colspan="2">CHEM1043 General chemistry II (6 credits)</td><td>Academic Year</td><td>2025</td></tr>
<tr><td>Offering Department</td><td>Chemistry</td><td>Quota</td><td>280</td></tr>
<tr><td>Course Co-ordinator</td><td colspan="3">Dr A T L Li, Chemistry < litla@hku.hk ></td></tr>
<tr><td>Teachers Involved</td><td colspan="3">(Dr A T L Li,Chemistry)
(Prof D L Phillips,Chemistry)
(Prof S K Lee,Chemistry)</td></tr>
<tr><td>Course Objectives</td><td colspan="3">This course is a continuation of CHEM1042 General Chemistry I. It aims to further consolidate some of the important fundamentals of chemistry that underlie many topics and principles across the physical sciences. The course prepares students to pursue a major in chemistry or in other aspects that require a good foundation in chemistry.</td></tr>
<tr><td>Course Contents & Topics</td><td colspan="3">1. Gases
Simple gas laws; ideal gas equation; gases in chemical reactions; mixture of gases; kinetic-molecular theory of gases; diffusion and effusion; non-ideal gases.

2. Structure and Bonding: The Delocalized Approach: Molecular Orbital Theory
Bonding in homonuclear and heteronuclear diatomic molecules of first and second period of elements; bonding in some simple polyatomic molecules; bonding in metals (band theory).

3. Solutions and their Properties
Types of solutions; intermolecular forces and the solution process; solution formation and equilibrium; solubilities of gases; vapor pressures of solutions; osmotic pressure; freezing-point depression and boiling-point elevation of nonelectrolyte solutions; solutions of electrolytes; colloidal mixtures.

4. Solubility and Complex-Ion Equilibria
Solubility product constant; relationship between solubility and Ksp; common-ion effect in solubility equilibria; limitations of the Ksp concept; precipitation; solubility and pH; equilibria involving complex ions; qualitative cation analysis.

5. Spontaneous Change
Entropy change; second law and third law of thermodynamics; Gibbs free energy change of a system; Gibbs free energy of reactions; coupled reactions

6. Electrochemistry
Electrode potentials and their measurement; standard electrode potentials; Ecell, delta G, and K; Ecell as a functions of concentrations; batteries; corrosion; electrolysis; industrial electrolysis processes.</td></tr>
<tr><td>Course Learning Outcomes</td><td colspan="3">On successful completion of this course, students should be able to:
<table><tr><td>CLO 1</td><td>demonstrate a knowledge and understanding of the properties and behavior of gases and apply gas laws and kinetic-molecular theory to processes involving gases</td></tr><tr><td>CLO 2</td><td>demonstrate a knowledge and understanding in relation to solutions and their properties, solubility and complex-ion equilibria, and also electrochemistry</td></tr><tr><td>CLO 3</td><td>apply molecular orbital theory to explain the formation and properties of diatomic molecules of first and second period of elements and of some simple polyatomic molecules</td></tr><tr><td>CLO 4</td><td>demonstrate a knowledge and understanding of the relationship between free energy and spontaneity of reaction</td></tr><tr><td>CLO 5</td><td>apply the theories and concepts introduced in the course to solve problems, perform calculations, make predictions and rationalize trends</td></tr><tr><td>CLO 6</td><td>organize and present chemical ideas in a clear, logical and coherent way</td></tr><tr><td>CLO 7</td><td>demonstrate awareness of the relevant applications of chemistry in society and in everyday life</td></tr></table></td></tr>
<tr><td>Pre-requisites
(and Co-requisites and Impermissible combinations)</td><td colspan="3">Pass in CHEM1042</td></tr>
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Course Status with Related Major/Minor /Professional Core	2025 Major in Biochemistry (Core/Compulsory) 2025 Major in Biological Sciences (Intensive) (Core/Compulsory) 2025 Major in Chemistry (Core/Compulsory) 2025 Major in Chemistry (Intensive) (Core/Compulsory) 2025 Major in Molecular Biology & Biotechnology (Intensive) (Core/Compulsory) 2025 Minor in Chemistry (Core/Compulsory) 2024 Major in Biochemistry (Core/Compulsory) 2024 Major in Biological Sciences (Intensive) (Core/Compulsory) 2024 Major in Chemistry (Core/Compulsory) 2024 Major in Chemistry (Intensive) (Core/Compulsory) 2024 Major in Molecular Biology & Biotechnology (Intensive) (Core/Compulsory) 2024 Minor in Chemistry (Core/Compulsory) 2023 Major in Biochemistry (Core/Compulsory) 2023 Major in Biological Sciences (Intensive) (Core/Compulsory) 2023 Major in Chemistry (Core/Compulsory) 2023 Major in Chemistry (Intensive) (Core/Compulsory) 2023 Major in Molecular Biology & Biotechnology (Intensive) (Core/Compulsory) 2023 Minor in Chemistry (Core/Compulsory) 2022 Major in Biochemistry (Core/Compulsory) 2022 Major in Biological Sciences (Intensive) (Core/Compulsory) 2022 Major in Chemistry (Core/Compulsory) 2022 Major in Chemistry (Intensive) (Core/Compulsory) 2022 Major in Molecular Biology & Biotechnology (Intensive) (Core/Compulsory) 2022 Minor in Chemistry (Core/Compulsory) 2021 Major in Biochemistry (Core/Compulsory) 2021 Major in Biological Sciences (Intensive) (Core/Compulsory) 2021 Major in Chemistry (Core/Compulsory) 2021 Major in Chemistry (Intensive) (Core/Compulsory) 2021 Major in Molecular Biology & Biotechnology (Intensive) (Core/Compulsory) 2021 Minor in Chemistry (Core/Compulsory)
Course to PLO Mapping	2025 Major in Biochemistry < PLO 1,2,3,4,5 > 2025 Major in Biological Sciences (Intensive) < PLO 1,2 > 2025 Major in Chemistry < PLO 1,2,4,5 > 2025 Major in Chemistry (Intensive) < PLO 1,2,4,5 > 2025 Major in Molecular Biology & Biotechnology (Intensive) < PLO 1,3,4,5 > 2024 Major in Biochemistry < PLO 1,2,3,4,5 > 2024 Major in Biological Sciences (Intensive) < PLO 1,2 > 2024 Major in Chemistry < PLO 1,2,4,5 > 2024 Major in Chemistry (Intensive) < PLO 1,2,4,5 > 2024 Major in Molecular Biology & Biotechnology (Intensive) < PLO 1,3,4,5 > 2023 Major in Biochemistry < PLO 1,2,3,4,5 > 2023 Major in Biological Sciences (Intensive) < PLO 1,2 > 2023 Major in Chemistry < PLO 1,2,4,5 > 2023 Major in Chemistry (Intensive) < PLO 1,2,4,5 > 2023 Major in Molecular Biology & Biotechnology (Intensive) < PLO 1,3,4,5 > 2022 Major in Biochemistry < PLO 1,2,3,4,5 > 2022 Major in Biological Sciences (Intensive) < PLO 1,2 > 2022 Major in Chemistry < PLO 1,2,4,5 > 2022 Major in Chemistry (Intensive) < PLO 1,2,4,5 > 2022 Major in Molecular Biology & Biotechnology (Intensive) < PLO 1,3,4,5 > 2021 Major in Biochemistry < PLO 1,2,3,4,5 > 2021 Major in Biological Sciences (Intensive) < PLO 1,2 > 2021 Major in Chemistry < PLO 1,2,4,5 > 2021 Major in Chemistry (Intensive) < PLO 1,2,4,5 > 2021 Major in Molecular Biology & Biotechnology (Intensive) < PLO 1,3,4,5 >
Offer in 2025 - 2026	Y 1st sem 2nd sem | Examination | Dec May
Offer in 2026 - 2027	Y
Course Grade	A+ to F
Grade Descriptors	A: Demonstrate thorough mastery at an advanced level of extensive knowledge and skills required for attaining all the course learning outcomes. Show thorough grasp of the subject. Demonstrate strong analytical and critical abilities and logical thinking, with ability to apply knowledge to a wide range of complex, familiar and unfamiliar situations. Apply highly effective organizational and presentational skills. B: Demonstrate substantial command of a broad range of knowledge and skills required for attaining at least most of the course learning outcomes. Show substantial grasp of the subject. Demonstrate evidence of analytical and critical abilities and logical thinking, and ability to apply knowledge to familiar and some unfamiliar situations. Apply effective organizational and presentational skills. C: Demonstrate general but incomplete command of knowledge and skills required for attaining most of the course learning outcomes. Show general but incomplete grasp of the subject. Demonstrate evidence of some analytical and critical abilities and logical thinking, and ability to apply knowledge to most familiar situations. Apply moderately effective organizational and presentational skills. D: Demonstrate partial but limited command of knowledge and skills required for attaining some of the course learning outcomes. Show partial but limited grasp, with retention of some relevant information, of the subject. Demonstrate evidence of some coherent and logical thinking, but with limited analytical and critical abilities. Show limited ability to apply knowledge to solve problems. Apply limited or barely effective organizational and presentational skills. Fail: Demonstrate little or no evidence of command of knowledge and skills required for attaining the course learning outcomes. Show evidence of little or no grasp of the knowledge and understanding of the subject. Lack of analytical and critical abilities, logical and coherent thinking. Show very little or no ability to apply knowledge to solve problems. Organization and presentational skills are minimally effective or ineffective.
Communication-intensive Course	N
Course Type	Lecture-based course

Course Teaching & Learning Activities

Activities	Details	No. of Hours
Lectures		36.0
Tutorials		12.0
Reading / Self study		100.0

Assessment Methods and Weighting

Methods	Details	Weighting in final course grade (%)	Assessment Methods to CLO Mapping
Assignments		30.0	1,2,3,4,5,6,7
Examination		50.0	1,2,3,4,5,6,7
Test		20.0	1,2,3,4,5,6,7

Required/recommended reading and online materials

(Textbook for the course) Petrucci; Herring; Madura; Bissonnette: General Chemistry - Principles and Modern Applications, latest edition, Pearson.
(Other reference books) Brown; LeMay; Bursten; Murphy; Woodward; Stoltzfus: Chemistry - The Central Science, latest edition, Pearson.
Tro: Chemistry - A Molecular Approach, latest edition, Pearson.
Robinson; McMurry; Fay: Chemistry, latest edition, Pearson.

Course Website

Additional Course Information

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